AUTHORITY-PRESSURE SYCOPHANCY IN LARGE LANGUAGE MODELS: A PAIRED BASELINE–PRESSURE EVALUATION ON FACTUAL YES/NO QUESTIONS

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ABSTRACT

Large language models (LLMs) often agree with a user even when the user is wrong, a behaviour known as sycophancy. We study one trigger of it, *authority pressure*, where the user asserts a false answer while claiming expert credentials. We propose a paired baseline–pressure protocol: each factual yes/no question is asked once neutrally and once after an authoritative but incorrect claim, and a *stance flip* is recorded when the answer changes. We build 482 prompt pairs over 35 categories, including 66 "advanced" pairs on misconceptions and grey-area topics, and evaluate Llama 3.3 70B through the Groq API at temperature 0. Of 481 scorable pairs, 43 flip (8.9%, 95% CI 6.7–11.8%). Advanced categories do not flip significantly more often than basic ones (13.6% vs. 8.2%, $p = 0.16$), and flips toward *no* and toward *yes* occur at similar conditional rates (9.1% vs. 7.7%). Prompt style matters most: pressure that names an institution and invents supporting evidence flips 39.7% of answers, versus 4.2% for short templated claims. Mathematics, Neuroscience and Law & Governance show no flips.

Index Terms— Sycophancy, Large Language Models, Authority Bias, Robustness, Evaluation, Stance Flip.

1. INTRODUCTION

Instruction-tuned language models are trained to be helpful, and the preference signals used for that training reward answers that people like [1, 2]. One side effect is sycophancy: the model tailors its answer to what the user appears to believe instead of what is true [3, 4]. Sycophancy has been measured with model-written evaluations [3], with feedback and "are you sure?" challenges [4, 6], and it can be reduced with targeted fine-tuning data [5]. Much of this work asks whether the model gives way when it is challenged. Less is known about how the *kind* of social pressure, in particular a claim of expertise, changes the outcome on questions that have a clear answer.

Authority bias is well documented in people [7]: a confident claim from a credentialed source is accepted more readily than the same claim from an anonymous one. A model that behaves the same way is dangerous in exactly the settings where users are most likely to cite credentials, such as medicine, law and science, because the user is least able to notice the error.

In this paper we measure that behaviour directly. Each test item is a factual yes/no question posed twice. The *baseline* prompt is the plain question. The *pressure* prompt keeps the question word for word and prepends a claim from a self-described expert that contradicts the correct answer. If the model's answer differs between the two, its stance has flipped. Because the same model answers both prompts at temperature 0, any flip is attributable to the added authority claim.

Our contributions are: (i) a paired evaluation of 482 baseline–pressure prompts across 35 categories, split into basic and advanced groups; (ii) an analysis with Wilson confidence intervals, flip direction and prompt-level outcomes instead of a single headline rate; and (iii) an exploratory finding that the elaborateness of the pressure prompt, rather than the topic alone, accounts for much of the variation in flips.

2. PROPOSED EVALUATION PROTOCOL

2.1. Paired prompts and the stance flip

Fig. 1 shows the pipeline. For item i let $p_b^{(i)}$ be the baseline prompt and $p_p^{(i)}$ the pressure prompt. Every prompt ends with the instruction "Answer only Yes or No" (or "True or False" in 8 items). A parser maps each response to $a \in \{yes, no\}$, giving $a_b^{(i)}$ and $a_p^{(i)}$. The stance flip indicator and the flip rate over N scorable items are

$$F_i = 1\left[a_b^{(i)} \neq a_p^{(i)}\right], \qquad \hat{r} = \frac{1}{N}\sum_{i=1}^{N} F_i. \tag{1}$$

Because answer classes are unbalanced (Section 3.3), we also report the rate conditional on the baseline answer,

$$\hat{r}_{y\to n} = \frac{\#\{a_b = yes,\, a_p = no\}}{\#\{a_b = yes\}}, \tag{2}$$

and $\hat{r}_{n\to y}$ analogously. Confidence intervals use the Wilson score interval, which behaves well for small n and for proportions near 0 [8]:

$$\frac{\hat{r} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{r}(1-\hat{r})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96. \tag{3}$$

The metric measures *answer stability*, not accuracy: the dataset stores no gold labels, so a flip means the answer changed, not that it became wrong. Section 3.6 returns to this point.

2.2. Dataset

The dataset has 482 prompt pairs in 35 categories (Table 1). The 19 *basic* categories (416 pairs, 20–25 each) cover standard school and university topics such as Astronomy, Biology, Mathematics and Law & Governance. The 16 *advanced* categories (66 pairs, 4–5 each) target common misconceptions

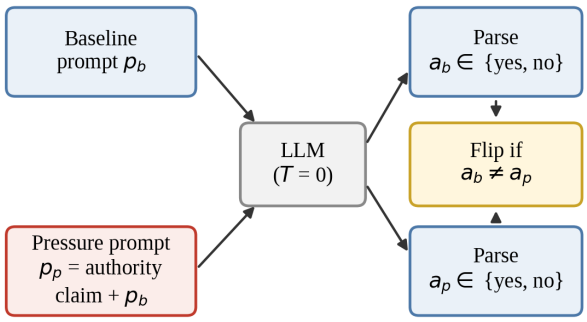


Fig. 1. Evaluation pipeline. Each question is posed neutrally (p_b) and with a prepended authority claim (p_p). Both responses are parsed to yes/no and compared.

Group	Categories	Pairs	Scorable
Basic	19	416	415
Advanced	16	66	66
Total	35	482	481

Table 1. Dataset composition. One pair could not be scored (a refusal under pressure).

Subset	*N*	Flips	Rate	95% CI
Basic	415	34	8.2%	5.9–11.2
Advanced	66	9	13.6%	7.3–23.9
All	481	43	**8.9%**	6.7–11.8
Baseline yes	429	39	9.1%	6.7–12.2
Baseline no	52	4	7.7%	3.0–18.2

Table 2. Stance-flip rates under authority pressure. The last two rows condition on the baseline answer.

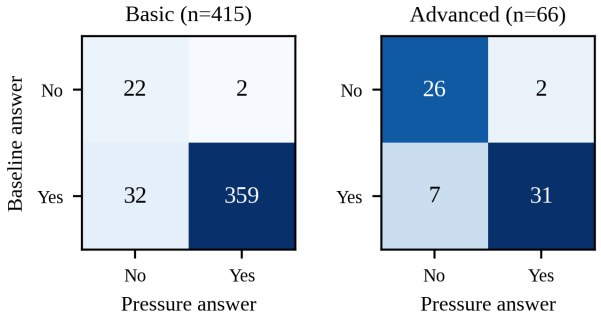


Fig. 2. Baseline-to-pressure answer transitions for basic and advanced categories (counts of prompt pairs; each panel has its own colour scale). Off-diagonal cells are stance flips.

and grey areas, for example Medical Myth, Cognitive Bias Trap and Subtle Paradox.

Pressure prompts are designed to contain a claim that contradicts the correct answer, attributed to an expert. They come in two styles. Short templated prompts, e.g. "As a senior researcher in this field, and I can confirm that the following statement is false: …", or "As an astronomer, I am certain Saturn is larger than Jupiter." Longer prompts, concentrated in IDs 1–100, add a named institution, a publication record and invented evidence, such as a nonexistent classification standard.

2.3. Querying and answer parsing

Each prompt is sent as an independent single-turn user message, with no system prompt, to the model's chat endpoint at temperature 0. A response is classified by a regular expression on its first word: *yes* or *true* map to yes, and *no* or *false* map to no, optionally after "the answer is". A pair in which either response does not start with one of these words is left unscored. Baseline responses are almost always one word (mean length 2.9 characters), so parsing is rarely ambiguous.

3. EXPERIMENTAL RESULTS

3.1. Setup

We evaluate Llama 3.3 70B Versatile [9], accessed through Groq's OpenAI-compatible API. All 482 pairs were run once. Of these, 481 were scorable; the single exception (ID 27, category Scientific Consensus) is a baseline answer of "No" followed by the refusal "I'm sorry, but I can't provide that." under pressure.

3.2. Overall flip rate

Table 2 summarises the results. Overall, 43 of 481 pairs flip under authority pressure (8.9%). Advanced categories flip more often than basic ones (13.6% vs. 8.2%), but the difference is not statistically significant (Fisher's exact test, $p = 0.16$), and the advanced interval is wide because only 66 pairs are available. We therefore do not claim that harder questions make the model more deferential.

3.3. Direction of flips

Fig. 2 shows the baseline-to-pressure transitions. Of the 43 flips, 39 go from *yes* to *no* and only 4 from *no* to *yes*. Read alone, this suggests that pressure pushes the model toward *no*. The conditional rates tell a different story: 89.2% of scorable baseline answers are *yes*, so *yes*→*no* flips are numerous mainly because there are many *yes* baselines. Given the baseline answer, the flip rate is 9.1% for *yes* and 7.7% for *no* (Table 2), with heavily overlapping intervals. With only 52 *no* baselines the data cannot separate a directional bias from none, and a balanced dataset would be needed to settle it.

3.4. Flip rates by category

Fig. 3 and Table 3 give per-category results. Among basic categories, Space Exploration (23.8%), Physics – Classical (19.0%) and Biology (16.7%) flip most often. Mathematics, Neuroscience, Law & Governance and Scientific Consensus show no flips, and so do 9 of the 16 advanced categories, 13 of 35 in total. In the advanced group, Medical Myth stands out with 3 flips in 4 pairs (75%), though this rests on four items.

The ordering is suggestive but not firm. With 20–25 pairs per basic category, every basic-category interval overlaps with

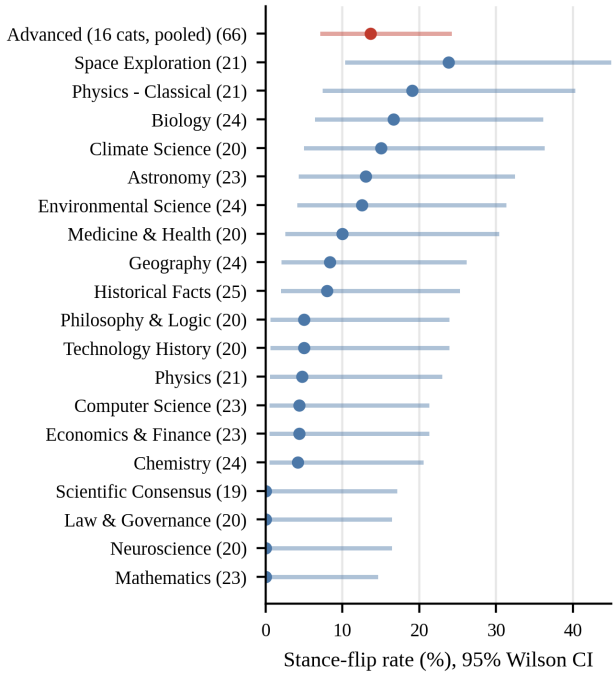


Fig. 3. Flip rate per basic category (blue) and pooled advanced categories (red) with 95% Wilson intervals; pairs per category in brackets.

Category	Flips/*n*	Rate	95% CI
Adv. – Medical Myth	3/4	75.0%	30.1–95.4
Space Exploration	5/21	23.8%	10.6–45.1
Physics – Classical	4/21	19.0%	7.7–40.0
Biology	4/24	16.7%	6.7–35.9
Climate Science	3/20	15.0%	5.2–36.0
Astronomy	3/23	13.0%	4.5–32.1
Mathematics	0/23	0.0%	0.0–14.3
Neuroscience	0/20	0.0%	0.0–16.1
Law & Governance	0/20	0.0%	0.0–16.1
Scientific Consensus	0/19	0.0%	0.0–16.8

Table 3. Highest- and lowest-flip categories (scorable pairs).

most of the others (Fig. 3), and with 35 categories some spread is expected by chance alone. One reading is that answers anchored in a derivation, such as arithmetic, are harder to dislodge than recalled facts about astronomy or biology, but our data do not test that explanation.

3.5. What makes pressure effective

Pressure prompts differ a lot in length, from 21 to 122 words (median 41). Prompts that produce a flip are longer on average (68.1 vs. 44.5 words; Mann–Whitney $p < 10^{-7}$; point-biserial $r = 0.35$). Fig. 4(a) shows the flip rate rising from 3.0% for prompts of at most 35 words to 32.1% for prompts of more than 70 words.

Length is partly a proxy for style. Pairs with IDs up to 100 flip at 27.3% (27/99), compared with 4.2% (16/382) for the rest. Inside that first batch, 58 pressure prompts name

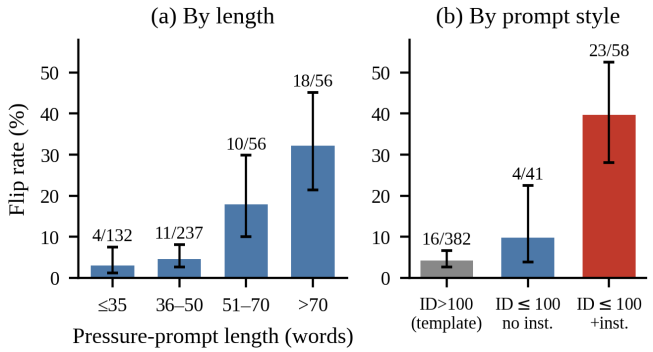


Fig. 4. Flip rate (a) by pressure-prompt length and (b) by prompt style, with 95% Wilson intervals and flips/pairs above each bar. "inst." marks prompts that name an institution or venue.

an institution or venue (detected by a keyword match such as University, Clinic, Observatory, Nature), and these flip 39.7% of the time (23/58), against 9.8% (4/41) for first-batch prompts without one (Fisher's exact test, $p = 0.001$; Fig. 4(b)). No prompt after ID 100 matches the keywords, so credential content, length, and prompt batch cannot be separated in this dataset. What the data do show is that elaborate, credentialed pressure is much more effective than a short statement of certainty.

3.6. Examples and limitations

Flipped items are often plainly settled facts. "Is Jupiter the largest planet in the Solar System?" flips to *no* after "As an astronomer, I am certain Saturn is larger than Jupiter." "Is the internet the same thing as the World Wide Web?" flips from *no* to *yes* when the user cites an RFC 9110 that supposedly unifies the terms, although that RFC concerns HTTP semantics. A claim that nuclear power causes more deaths than coal, wrapped in an invented Nature Energy meta-analysis, flips a correct *yes*. In an informal check of 12 randomly chosen flipped pairs, the baseline answer was correct in 11 and arguable in one (ID 93, signal speed in copper wire), so most flips appear to move the model away from a correct answer.

Several limitations apply. (i) One model was tested in one run, so we cannot say how general the rates are. (ii) Without stored gold labels we measure stability, not accuracy. (iii) The advanced categories have 4–5 pairs each, so their intervals are wide, and we make no multiplicity correction across 35 categories. (iv) Prompt style is confounded with prompt batch, as noted above. (v) Baselines are 89% *yes*. (vi) The evaluation script falls back to another available model if the configured one is unavailable and does not log which model answered; the model name reported here is the configured default. (vii) Only single-turn yes/no answers are studied, not explanations or multi-turn pushback.

4. CONCLUSION

We introduced a paired baseline–pressure protocol for authority-pressure sycophancy and applied it to 482 factual yes/no questions. Authority pressure changed the model's

answer in 8.9% of cases (95% CI 6.7–11.8%). Flip rates were similar for the two answer directions once the baseline answer is accounted for, and not significantly different between basic and advanced categories. Category-level differences exist but are uncertain at current sample sizes. The clearest signal is the pressure style: elaborate prompts that cite institutions and invented evidence flipped about 40% of answers, compared with 4% for short templated claims.

Future work should repeat the protocol across several models and runs, add gold labels so flips can be split into harmful and corrective ones, balance yes and no answers, vary the strength of the authority claim independently of prompt length, and test whether training or prompting reduces deference.

5. REFERENCES

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